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Data Pipeline with DuckDB + DBT + Airflow 3 PoC with theory.

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Nazar Khimin Follow

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Hello, I would like to share my recent experience of writing simple ETL and describe used tools and approaches.

I forked: https://github.com/dbt-labs/jaffle_shop_duckdb — playground dbt project where I added:

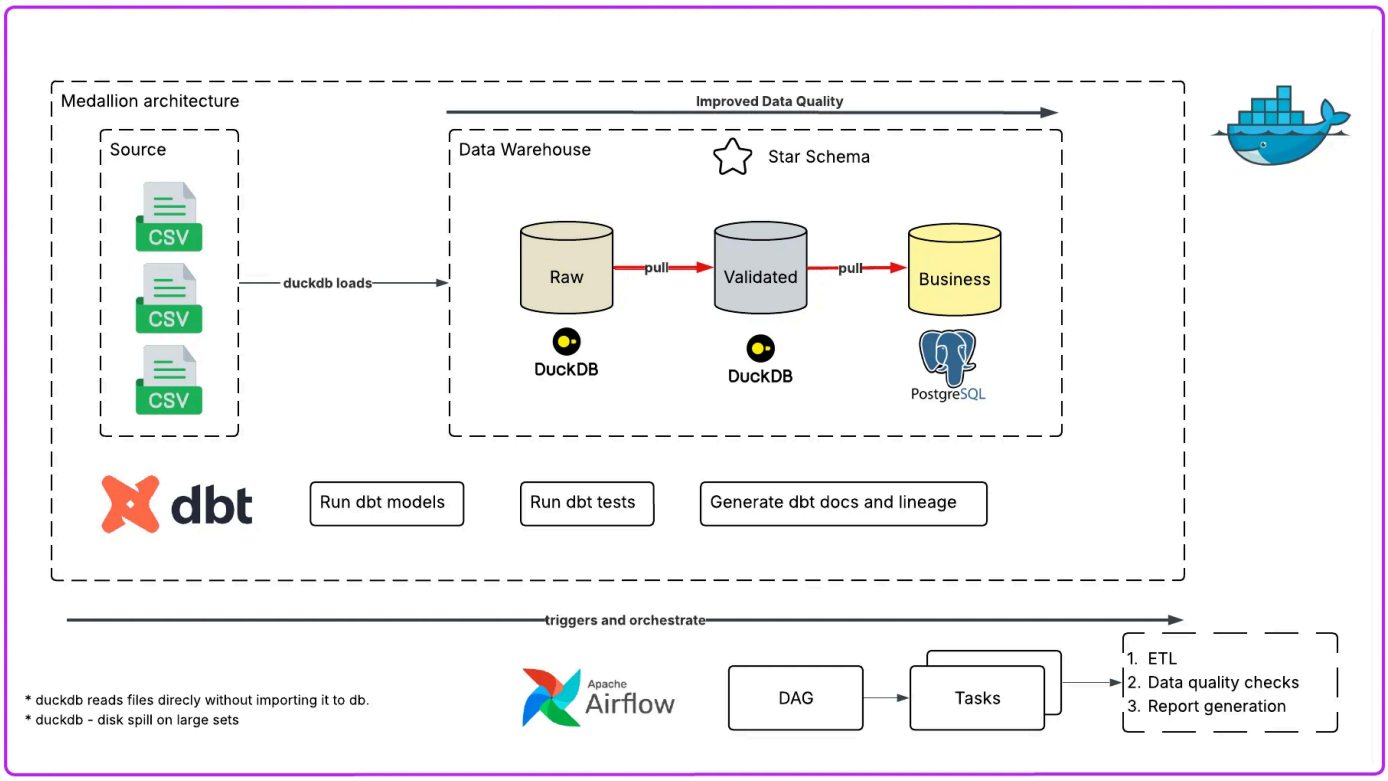
- ✅ **Loading CSV files using DuckDB** (no `dbt seed`)
- 🧱 **Medallion Architecture** applied for data transformation (bronze → silver → gold)
- 🚀 **Gold layer** loaded into **PostgreSQL**
- ⚙️ **Airflow DAG** that orchestrates `dbt run` and `dbt test`
- 📦 **Containerized** with Docker Compose
- 🔄 **Upgraded Airflow to v3.0.0**

Final working project here:

https://github.com/nazar-khimin/data_misc/tree/main/airflow/pii

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T - Transfor



Proposed materialization

Layer	Dev	Prod
Bronze	table	incremental
Silver	ephemeral	incremental
Bronze	table	incremental

Conext & Component diagram

🗂️ Agenda

We'll cover both the technical implementation and theory:

1. **Data engineering & Data pipelines**
2. **ETL vs ELT in modern data stacks**
3. **Medallion vs Data Mesh architecture**
4. **Overview of data processing tools:** DuckDB, Pandas, Polars, Spark
5. **OLTP vs OLAP**
6. **Power of DBT for data modeling**

7. Airflow as the orchestrator

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1. Data en

Data engineers are responsible for transforming raw data into a format that data scientists, analysts, and other professionals can use for actionable insights.

Think of data engineering as the backbone of any data-driven operation — it ensures that data is available, accurate, and ready for analysis.

It allows organizations to efficiently manage large volumes of data and extract valuable insights that inform strategic decisions.

Data Pipelines — The Heart of Data Engineering

Pipelines are automated processes that extract raw data from its source, clean and transform it, and deliver it to a destination for storage and analysis. They are essential because they enable continuous and efficient data flow throughout an organization.

Types of data pipelines:

1. **Batch processing**: These pipelines process data in large chunks at scheduled intervals.
2. **Streaming pipelines**: In contrast, streaming pipelines process data in real time.
3. **ETL (Extract, Transform, Load)**: ETL is a type of pipeline that first extracts data from its source, transforms it into a usable format, and the loads it into a data warehouse or other storage solution.
4. **ELT (Extract, Load, Transform)**: ELT is a newer approach where data is first extracted from its source and loaded directly into a data warehouse.

Data engineering lifecycle:

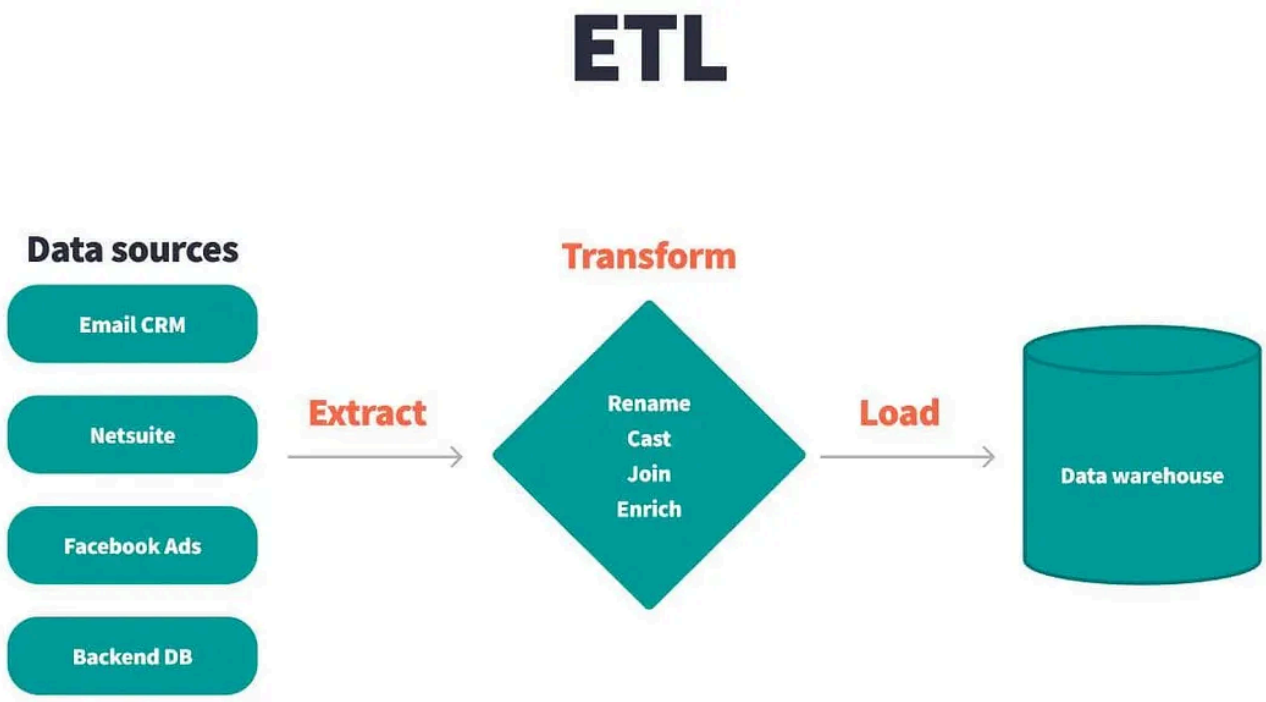
1. Data collection
2. Data storage
3. Data transformation
4. Data analysis
5. Data governance

Source: https://www.getdbt.com/blog/what-is-data-engineering

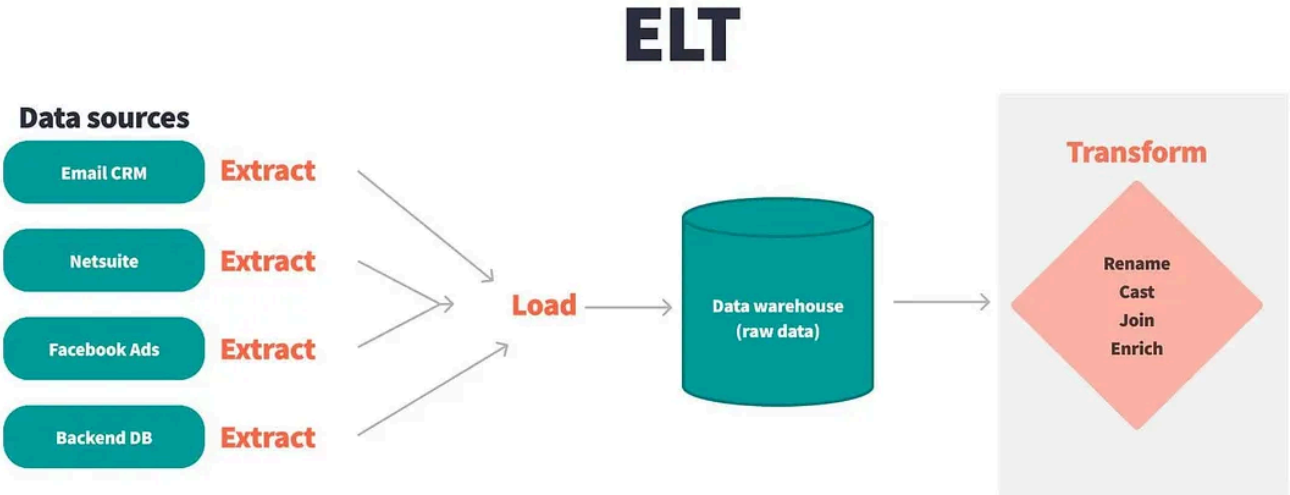
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2. ETL vs

ELT



ELT



This hybrid model offers the best of both worlds — maintaining data governance and control while leveraging cloud-based scalability for certain datasets.

ELT vs ETL · Pros and Cons

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Feature		
Transform location	Before loading	Inside destination
Speed (big data)	Slower	Faster with scalable systems
Flexibility	Less (transformed only once)	More (raw data remains)
Tools	Talend, Informatica, SSIS	dbt, Fivetran, Airbyte, Spark
Best for	Regulated, legacy systems	Cloud-native, agile workflows

3. Medallion vs Data Mesh Architecture

Medallion Architecture (Bronze → Silver → Gold):

- Bronze: Raw ingested data
- Silver: Cleaned and joined datasets
- Gold: Aggregated, business-ready tables

Data Mesh focuses more on **domain-driven** data ownership and decentralization.

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This PoC uses **Medallion** because it's intuitive, widely adopted, and aligns well with dbt's modular design.

4. Overview of data processing tools: DuckDB, Pandas, Polars, Spark

- **DuckDB:** Embedded SQL database for fast analytics. Works great with Parque
- **Pandas.** The go-to tool for Python data wrangling. Great for small to medium datasets.
- **Polars:** A faster, more memory-efficient alternative to Pandas with parallel execution.
- **Spark:** Distributed engine used in big data scenarios. Overhead is high for small jobs.

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DuckDB hits a sweet spot: SQL + speed + simplicity.

5. OLTP vs OLAP

- **OLTP (Online Transaction Processing):** Used in production systems. Fast inserts, updates, deletes (e.g., PostgreSQL for app backend).
- **OLAP (Online Analytical Processing):** Optimized for analytical queries over large datasets (e.g., DuckDB, BigQuery).

Category	OLTP	OLAP
Purpose	Handle transactions	Analyze data
Operations	Insert, Update, Delete	Complex Queries, Aggregations
Speed	Fast for individual records	Fast for large scans
Data	Real-time, current state	Historical, bulk data
Example Tools	PostgreSQL, MySQL	DuckDB, Redshift, Snowflake

Our ETL pipeline is focused on **OLAP**: we extract from files, transform with DuckDB, and push aggregates to OLTP (PostgreSQL) only if needed.

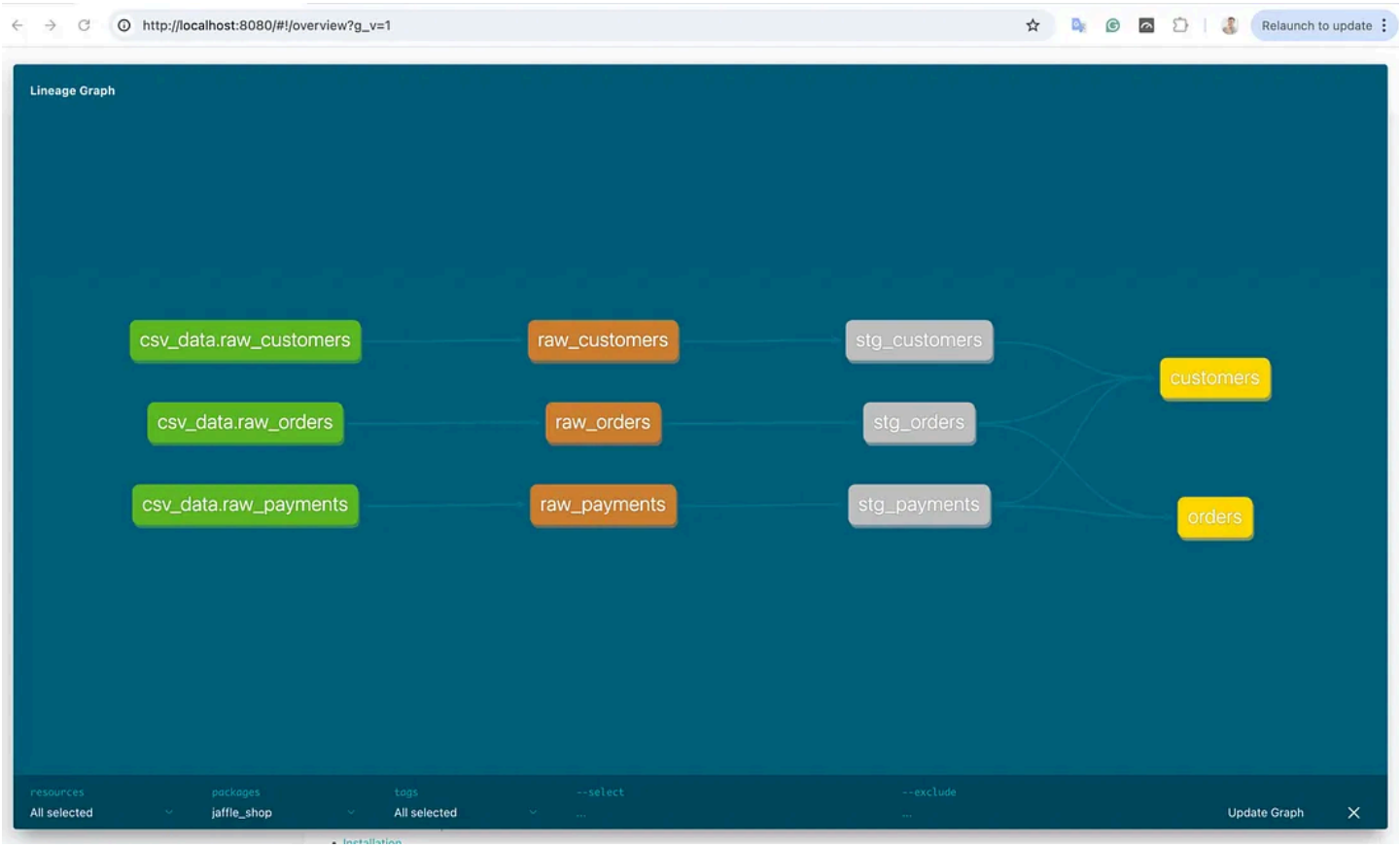
💡 **Takeaway:**

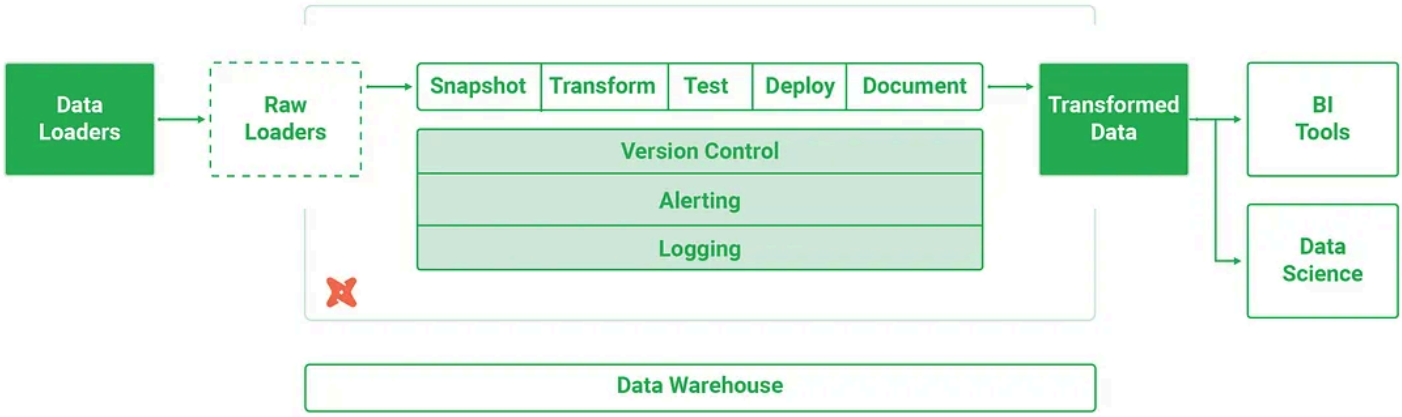
ETL/ELT pipelines serve as a bridge between **OLTP (operations)** and **OLAP (analytics)**.

6. Power of DBT (Data Build Tool)

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- Declar
- Data testing and documentation out of the box
- Modular DAGs: each model is a node in the graph
- Excellent integration with version control





7. Airflow as the orchestrator

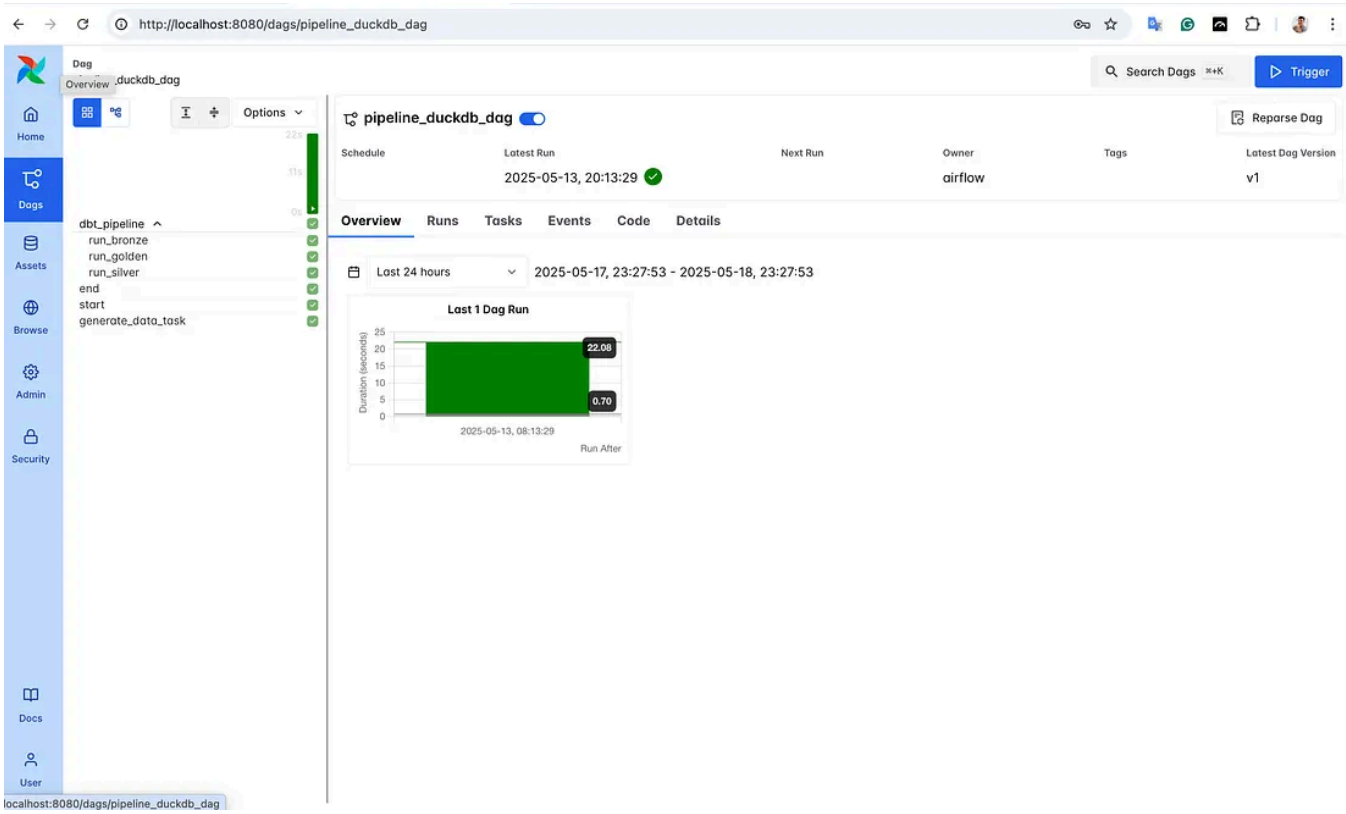
Apache Airflow is an open-source tool used to **orchestrate workflows** — meaning it lets you de

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on code.

- Airflow is a **workflow orchestration platform** and implement **workflow as code** approach
- It allows you to define workflows as **Directed Acyclic Graphs (DAGs).**
- Each DAG represents a data pipeline — a set of tasks with dependencies and order of execution.

💡 Think of Airflow as the **"conductor"** of your data workflows. It ensures each task runs in the right order, at the right time, and gets retried if it fails.



DAGs

A DAG is a model that encapsulates everything needed to execute a workflow. Some DAG attributes include the following:

- **Schedule**: When the workflow should run.
- **Tasks:** tasks are discrete units of work that are run on workers.
- **Task Dependencies**: The order and conditions under which tasks execute.
- **Callbacks:** Actions to take when the entire workflow completes.

- **Additional Parameters:** And many other operational details

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Airflow® is tended for continuously running, event-driven, or streaming workloads



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